Slide 1 · Cellular respiration overview

Glucose + 6 O2 → 6 CO2 + 6 H2O + energy (ATP). Three stages: glycolysis, the Krebs cycle, the electron transport chain. Covered on the midterm.

Slide 2 · Glycolysis

Happens in the cytosol. One glucose (6C) becomes two pyruvate (3C). Net gain: 2 ATP and 2 NADH. Does not need oxygen.

Slide 3 · Krebs cycle

In the mitochondrial matrix. Acetyl-CoA enters; per glucose: 2 ATP, 6 NADH, 2 FADH2, 4 CO2 released.

Slide 4 · Electron transport chain

Inner mitochondrial membrane. NADH and FADH2 donate electrons; oxygen is the final electron acceptor; chemiosmosis drives ATP synthase. About 26–28 ATP.

Slide 5 · Fermentation

Without oxygen: lactic acid fermentation (muscle) or alcohol fermentation (yeast) regenerates NAD+ so glycolysis can continue.